

Лабораторная работа

Номер 12

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01 января 1970

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Информация

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Получение навыков по управлению системным временем и настройке синхронизации времени

Вывод команды timedatectl на сервере

```
[nsandryushin@server.nsandryushin.net ~]$ timedatectl
      Local time: Thu 2025-11-20 20:34:10 UTC
     Universal time: Thu 2025-11-20 20:34:10 UTC
          RTC time: Thu 2025-11-20 20:34:10
        Time zone: UTC (UTC, +0000)
System clock synchronized: yes
          NTP service: active
    RTC in local TZ: no
```

Рис. 1: Вывод команды timedatectl на сервере

Просмотр свойств времени с помощью `timedatectl show`

```
[nsandryushin@server.nsandryushin.net ~]$ timedatectl show
Timezone=UTC
LocalRTC=no
CanNTP=yes
NTP=yes
NTPSynchronized=yes
TimeUSec=Thu 2025-11-20 20:36:10 UTC
RTCTimeUSec=Thu 2025-11-20 20:36:09 UTC
```

Рис. 2: Просмотр свойств времени с помощью `timedatectl show`

Вывод команды timedatectl на клиенте

```
[nsandryushin@client.nsandryushin.net ~]$ timedatectl
      Local time: Thu 2025-11-20 20:35:19 UTC
    Universal time: Thu 2025-11-20 20:35:19 UTC
          RTC time: Thu 2025-11-20 20:35:19
        Time zone: UTC (UTC, +0000)
System clock synchronized: yes
          NTP service: active
    RTC in local TZ: no
```

Рис. 3: Вывод команды timedatectl на клиенте

Эксперименты с параметрами команды date на сервере

```
[nsandryushin@server.nsandryushin.net ~]$ date
Thu Nov 20 08:36:28 PM UTC 2025
[nsandryushin@server.nsandryushin.net ~]$ date -dR
Thu Nov 20 05:00:00 AM UTC 2025
[nsandryushin@server.nsandryushin.net ~]$ date -d
date: option requires an argument -- 'd'
Try 'date --help' for more information.
[nsandryushin@server.nsandryushin.net ~]$ date --date=STRING
date: invalid date 'STRING'
[nsandryushin@server.nsandryushin.net ~]$ date -R
Thu, 20 Nov 2025 20:38:28 +0000
[nsandryushin@server.nsandryushin.net ~]$ date
Thu Nov 20 08:38:36 PM UTC 2025
[nsandryushin@server.nsandryushin.net ~]$ date -R
Thu, 20 Nov 2025 20:38:37 +0000
```

Рис. 4: Эксперименты с параметрами команды date на сервере


```
[nsandryushin@client.nsandryushin.net ~]$ date  
Thu Nov 20 08:36:38 PM UTC 2025
```

Рис. 5: Просмотр системного времени на клиенте

Просмотр аппаратного времени на сервере

```
[nsandryushin@server.nsandryushin.net ~]$ sudo hwclock  
[sudo] password for nsandryushin:  
2025-11-20 20:39:18.794191+00:00
```

Рис. 6: Просмотр аппаратного времени на сервере

Просмотр аппаратного времени на клиенте

```
[nsandryushin@client.nsandryushin.net ~]$ sudo hwclock  
[sudo] password for nsandryushin:  
2025-11-20 20:39:45.923811+00:00
```

Рис. 7: Просмотр аппаратного времени на клиенте

Просмотр источников времени на сервере

```
[nsandryushin@server.nsandryushin.net ~]$ chronyc sources
MS Name/IP address          Stratum Poll Reach LastRx Last sample
=====
^- 92.241.12.152             3    8   377   139   +130us[ +130us] +/-   73ms
^+ ntp1.mail.ru              2    8   377    77   +504us[ +504us] +/-  5844us
^* mskstd-ntp01c.ntppool.ya> 2    8   377   209    +48us[ +54us] +/-   11ms
^+ 192.36.143.130           1    8   377    25   +782us[ +782us] +/-   15ms
```

Рис. 8: Просмотр источников времени на сервере

Просмотр источников времени на клиенте

```
[nsandryushin@client.nsandryushin.net ~]$ chronyc sources
MS Name/IP address          Stratum Poll Reach LastRx Last sample
=====
^- 195.54.177.84              2   8   377   168  -1758us[-1869us] +/-   28ms
^+ mskstd-ntp01c.ntppool.ya>  2   6   377    50   +684us[ +684us] +/-  6458us
^* unspecified.mtw.ru         2   8   377   174  -2729us[-2840us] +/-  8471us
^+ vm2300563.firstbyte.club   2   8   377   112   -372us[ -372us] +/-  6680us
```

Рис. 9: Просмотр источников времени на клиенте

Настройка доступа к NTP-серверу для локальной сети

```
GNU nano 8.1 /etc/chrony.conf Modified
# Use public servers from the pool.ntp.org project.
# Please consider joining the pool (https://www.pool.ntp.org/join.html).
pool 2.rocky.pool.ntp.org iburst

# Use NTP servers from DHCP.
sourcedir /run/chrony-dhcp

# Record the rate at which the system clock gains/losses time.
driftfile /var/lib/chrony/drift

# Allow the system clock to be stepped in the first three updates
# if its offset is larger than 1 second.
makestep 1.0 3

allow 192.168.0.0/16
```

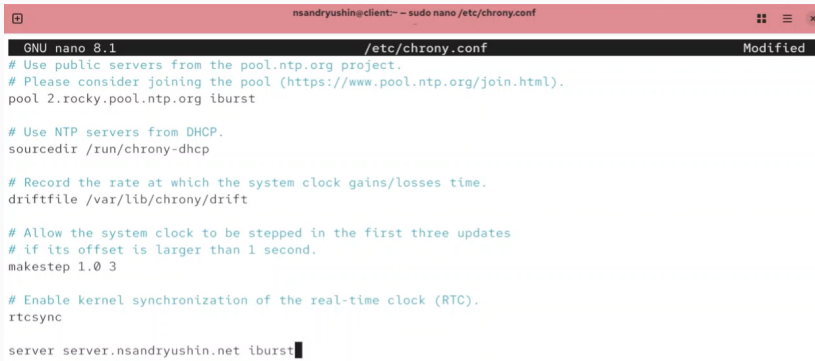
Рис. 10: Настройка доступа к NTP-серверу для локальной сети

Перезапуск службы и настройка Firewall на сервере

```
[nsandryushin@server.nsandryushin.net ~]$ systemctl restart chronyd  
[nsandryushin@server.nsandryushin.net ~]$ firewall-cmd --add-service=ntp --permanent  
success  
[nsandryushin@server.nsandryushin.net ~]$ firewall-cmd --reload  
success
```

Рис. 11: Перезапуск службы и настройка Firewall на сервере

Настройка клиента на использование локального NTP-сервера



```
nsandryushin@client:~ - sudo nano /etc/chrony.conf
GNU nano 8.1 /etc/chrony.conf Modified
# Use public servers from the pool.ntp.org project.
# Please consider joining the pool (https://www.pool.ntp.org/join.html).
pool 2.rocky.pool.ntp.org iburst

# Use NTP servers from DHCP.
sourcedir /run/chrony-dhcp

# Record the rate at which the system clock gains/losses time.
driftfile /var/lib/chrony/drift

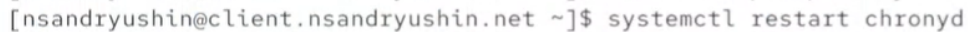
# Allow the system clock to be stepped in the first three updates
# if its offset is larger than 1 second.
makestep 1.0 3

# Enable kernel synchronization of the real-time clock (RTC).
rtcsync

server server.nsandryushin.net iburst
```

Рис. 12: Настройка клиента на использование локального NTP-сервера

Перезапуск службы chronyd на клиенте

A terminal window showing a command being executed. The prompt is [nsandryushin@client.nsandryushin.net ~]\$. The command is systemctl restart chronyd.

```
[nsandryushin@client.nsandryushin.net ~]$ systemctl restart chronyd
```

Рис. 13: Перезапуск службы chronyd на клиенте

Проверка источников времени на сервере

```
[nsandryushin@server.nsandryushin.net ~]$ chronyc sources
MS Name/IP address          Stratum Poll Reach LastRx Last sample
=====
^* 188.68.222.140             2    6    37    20    -608us[-6987us] +/-   30ms
^- 185.211.244.47             2    6    37    28    +9042us[+9042us] +/-   31ms
^- 62.76.113.232              3    6    37    30     -37ms[ -37ms] +/-   76ms
^- 85.193.65.152              2    6    37    30     +13ms[ +13ms] +/-   56ms
```

Рис. 14: Проверка источников времени на сервере

Проверка источников времени на клиенте

```
[nsandryushin@client.nsandryushin.net ~]$ chronyc sources
MS Name/IP address          Stratum Poll Reach LastRx Last sample
=====
^- 85.193.65.152             2    6    17    10    +890us[ +890us] +/-   52ms
^- 188.68.222.140           2    6    17    24    +800us[ +800us] +/-   22ms
^* 62.76.113.232            3    6    17    30   +2958us[+3482us] +/-   27ms
^- 185.211.244.47           2    6    17    29   -4440us[-4440us] +/-   36ms
^+ server.nsandryushin.net   3    6    17    29   -5979us[-5979us] +/-   35ms
```

Рис. 15: Проверка источников времени на клиенте

Просмотр детальной информации о синхронизации на сервере

```
[nsandryushin@server.nsandryushin.net ~]$ chronyc tracking
Reference ID      : BC44DE8C (188.68.222.140)
Stratum          : 3
Ref time (UTC)   : Thu Nov 20 20:46:19 2025
System time      : 0.000740243 seconds fast of NTP time
Last offset      : +0.000755800 seconds
RMS offset       : 0.006307106 seconds
Frequency        : 487.993 ppm fast
Residual freq    : +10.201 ppm
Skew             : 1.192 ppm
Root delay       : 0.033041012 seconds
Root dispersion  : 0.010776951 seconds
Update interval  : 64.8 seconds
Leap status      : Normal
```

Рис. 16: Просмотр детальной информации о синхронизации на сервере

```
[nsandryushin@server.nsandryushin.net ~]$ cd /vagrant/provision/server
[nsandryushin@server.nsandryushin.net server]$ sudo mkdir -p /vagrant/provision/server/ntp/etc
[sudo] password for nsandryushin:
[nsandryushin@server.nsandryushin.net server]$ sudo cp -R /etc/chrony.conf /vagrant/provision/server/ntp/
etc/
[nsandryushin@server.nsandryushin.net server]$ cd /vagrant/provision/server
[nsandryushin@server.nsandryushin.net server]$ touch ntp.sh
[nsandryushin@server.nsandryushin.net server]$ chmod +x ntp.sh
[nsandryushin@server.nsandryushin.net server]$ nano ntp.sh
[nsandryushin@server.nsandryushin.net server]$ █
```

Рис. 17: Vagrant

Содержимое скрипта настройки ntp.sh для сервера



The screenshot shows a terminal window with the title bar "nsandryushin@server:/vagrant/provision/server - nano ntp.sh". The terminal content is as follows:

```
GNU nano 8.1                                ntp.sh                                Modified
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y install chrony
echo "Copy configuration files"
cp -R /vagrant/provision/server/ntp/etc/* /etc
restorecon -vR /etc
echo "Configure firewall"
firewall-cmd --add-service=ntp
firewall-cmd --add-service=ntp --permanent
echo "Restart chronyd service"
systemctl restart chronyd
```

Рис. 18: Содержимое скрипта настройки ntp.sh для сервера

```
[nsandryushin@client.nsandryushin.net ~]$ cd /vagrant/provision/client
[nsandryushin@client.nsandryushin.net client]$ sudo mkdir -p /vagrant/provision/client/ntp/etc
[sudo] password for nsandryushin:
[nsandryushin@client.nsandryushin.net client]$ cp -R /etc/chrony.conf /vagrant/provision/client/ntp/etc/
[nsandryushin@client.nsandryushin.net client]$ cd /vagrant/provision/client
[nsandryushin@client.nsandryushin.net client]$ touch ntp.sh
[nsandryushin@client.nsandryushin.net client]$ chmod +x ntp.sh
[nsandryushin@client.nsandryushin.net client]$ nano ntp.sh
[nsandryushin@client.nsandryushin.net client]$ █
```

Рис. 19: Vagrant

Содержимое скрипта настройки ntp.sh для клиента



```
nsandryushin@client:/vagrant/provision/client - nano ntp.sh
/vagrant/provision/client
GNU nano 8.1 ntp.sh
#!/bin/bash
echo "Provisioning script $0"
echo "Copy configuration files"
cp -R /vagrant/provision/client/ntp/etc/* /etc
restorecon -vR /etc
echo "Restart chronyd service"
systemctl restart chronyd
```

Рис. 20: Содержимое скрипта настройки ntp.sh для клиента


```
109 server.vm.provision "server_ntp",
110                       type: "shell",
111                       preserve_order: true,
112                       path: "provision/server/ntp.sh"
113
114 end
115
116 ## Client configuration
117 config.vm.define "client", autostart: false do |client|
118   client.vm.box = "rockylinux8"
119   client.vm.hostname = "client"
120
121   client.vm.boot_timeout = 1440
122
123   client.ssh.insert_key = false
124   client.ssh.username = "vagrant"
125   client.ssh.password = "vagrant"
126
127   client.vm.network :private_network,
128                     type: "dhcp",
129                     virtualbox____innet: true
130
131   client.vm.provider :virtualbox do |virtualbox|
132     virtualbox.customize ["modifyvm", :id, "--vrdm", "on"]
133     virtualbox.customize ["modifyvm", :id, "--vrdmport", "3392"]
134   end
135
136   client.vm.provision "client_dummy",
137                       type: "shell",
138                       preserve_order: true,
139                       path: "provision/client/01-dummy.sh"
140
141   client.vm.provision "client_routing",
142                       type: "shell",
143                       preserve_order: true,
144                       run: "always",
145                       path: "provision/client/01-routing.sh"
146
147   client.vm.provision "client_mail",
148                       type: "shell",
149                       preserve_order: true,
150                       path: "provision/client/mail.sh"
151
152   client.vm.provision "client_ntp",
153                       type: "shell",
154                       preserve_order: true,
155                       path: "provision/client/ntp.sh"
156
157 end
```

Рис. 21: Vagrantfile

В результате выполнения лабораторной работы были получены навыки настройки системного времени и ntp синхронизации